

Finalising Survey Part III

A three-day roadmap — experiment runs and redaction. Every figure below was measured on 2026-08-19/20 against branch checking-w-Claude-Code, not estimated.

0 Where things actually stand

The one fact that reorders everything: the manuscript does not currently compile. pdf latex stops with a fatal error at line 1506 because prof_zeta.tex and prof_mech.tex are \input-ed but absent from report/. Supplying both from the campaign archives makes a full latexmk run exit 0 with **zero undefined references and zero undefined citations** across 53 pages. Nothing else in the document is structurally broken.

Asset	State	Evidence
Package tests	1409 pass, 0 fail	Pkg.test(), 6m59s
Documentation	clean, warnonly=false	docs/make.jl exit 0
Manuscript build	FAILS → clean once 2 files exist	53 pages, 0 undefined refs
Campaign exp1–9	current at HEAD	archived 2026-08-19 16:26
Campaign exp8	restored — first run ever	was excluded by a wrong note
Campaign exp12	first archive ever, 242 s	135-run grid after your edit
Campaign exp13	reproduces exactly	RDFO 0.9968, RGradCapped 1.0000
Campaign exp11	partial — 2 defects, see Day 1	CIFAR leg killed at 5.5 h
Provisional numbers	19 live markers in the .tex	author's own flag
Unproduced figures	6 compiled in, no producer in repo	audited across all .jl/.ipynb

Two content gaps carry real risk, because both concern experiments that now have data and no home in the manuscript:

- **Flat well.** Experiment 13 reproduces, but sec: res flat well, prop: flat well, tab: flat well and fig: flat well do not exist. There is no subsection to receive the numbers.
- **Drifted table.** tab: sums disagrees with the current notebook output on **all five rows**, and its third column (Sum Δ^2/M_k) is marked provisional on every row — it has never been computed, although radius_sums now returns it.

Mechanism	MS $\Sigma\Delta$	Current	MS $\Sigma\Delta^2$	Current
R-delta	2.6e120	1.172e59	overflow	3.530e117
R-step	7.98e2	8.017e2	1.60e3	1.898e3
R-DFO	8.28e0	5.900e0	1.22e0	1.430e0
R-grad capped	6.48e0	7.248e0	4.28e-1	1.430e0
R-grad uncapped	2.77e2	3.740e2	1.92e2	2.591e2

Part of the R-step and R-grad drift is the stalled-iteration trace fix landing (393→394, 539→540 iterations); the rest predates it.

Day 1 — Make it build, make the numbers current

Roughly 6.5 hours. The first task is thirty minutes and unblocks everything downstream; do it before anything else.

#	Task	Done when	Est
1.1	Unblock the build. Copy the two generated profile tables into report/ under the names the manuscript expects, and add a make target so the copy is never a manual step again. exp3_perf_profile_zeta.tex → prof_zeta.tex exp1_perf_profile_iter.tex → prof_mech.tex	<code>latexmk -pdf</code> exits 0, 53 pages, 0 undefined refs. <i>Already verified in scratch.</i>	30 m
1.2	Fix experiment 11's two defects. (a) World age. @eval Main using MLDatasets then calling Main.MLDatasets.MNIST(...) in the same frame cannot see the new methods. Both legs have <i>always</i> silently used randn. Wrap in Base.invokelatest. (b) Cost guard. $n \leq 2000$ admits ExactHessian at $n=1156$, where it costs 24.6 s/iter against BHHH's 0.08 — 308x, and 99.3% of the leg. Recalibrate on time, not on the n-by-n array.	Cold session loads real MNIST (28,28,60000). Train accuracy stops reading 0.499 on a 10-class problem.	60 m
1.3	Relaunch exp11, MNIST only , in the background while you redact. Default bhhh_study() to (:mnist,); the CIFAR leg is a measured 4.1 h and buys one more misspecified-network row.	Archive with a summary file (the killed run has none).	10 m + 2.7 h bg
1.4	Refresh tab: sums from the current notebook output, including the caption, which is now far more careful than the manuscript's about what a partial sum over a finite run does and does not establish.	All five rows match tab_sums.txt.	45 m
1.5	Sweep the provisional numbers in the regimes and inactivity subsections (lines ~1222–1412: eight markers). Each one is either refreshed from an archive or the claim is withdrawn.	Zero \provisional left in those two subsections.	3 h
gate	End-of-day gate. Full <code>latexmk</code> , read the log, commit.	Exit 0, 0 undefined refs, 0 undefined citations.	15 m

Day 2 — The two decisions only you can make

Roughly 6.5 hours. Both items are judgement calls with data now behind them.

2.1 The true_stop decision

This is the one that touches the paper's credibility. The first-ever exp12 archive shows the effect in its own numbers, under `true_stop = "none"`:

cell (RadiusProportional)	iters	samples	true norm g	verdict
LS-linear / RGrad	1	4	2.299	noise stop
LS-linear / RDFO	4	69	0.680	noise stop
Logistic / RDFO	55	220	0.258	noise stop
LS-linear / RDelta	16	214,114	8.7e-08	genuine
Logistic / RDelta	144	4,548,398	5.5e-08	genuine

RGrad reports success after **one iteration on four samples** with a true gradient norm of 2.3. Read as an efficiency result that is badly wrong. The table's own footer already names the mechanism — a rule that shrinks the radius fast enough satisfies its own stopping test on noise alone — but with the confirmation off, the run is still recorded as converged.

Important scope limit, and it is good news: no *archived* number is affected. Every other experiment is deterministic or full-batch, and the solver *rejects* any `true_stop` other than `:none` there. The exposure is confined to exp12, which had never run.

#	Task	Done when	Est
2.1	Re-run the three affected cells under <code>true_stop = :test</code> and decide the reporting convention. Recommendation: keep <code>:none</code> as the default (it reproduces the literature and that is defensible) but report the true gradient norm beside every sampled row, and say in the caption which mode produced each number. <code>:test</code> costs nothing per iteration; the extra total cost is only because the run no longer stops early.	Every sampled row in the paper states its mode.	2 h

2.2 The flat-well subsection

#	Task	Done when	Est
2.2	Write <code>sec: res flat well</code> from scratch: proposition, table, figure. The numbers are measured and stable — RDFO holds a ratio of 0.9968 across the full zeta range and RGradCapped is exact at 1.0000, reproduced across two runs three days apart. Note the range is two decades of zeta (0.5 to 50), not four.	Four new labels resolve; ratio > 1 on an RDFO row would refute the proposition and none does.	2.5 h
2.3	Fill the $\text{Sum } \Delta^2/M_k$ column. Re-run the sums notebook with <code>hessian_norm = true</code> so <code>radius_sums</code> returns it. This is the column Part II's summability statement is actually about, so leaving it provisional weakens the section it appears in.	Column populated; no provisional markers in tab: sums.	1 h
2.4	exp11's MNIST numbers land from Day 1. Update the mechanism-against-model subsection, and add one sentence recording that the network legs use synthetic data of the same shape unless the download succeeded — now true rather than accidental.	Provisional markers at ~1734 and ~1756 cleared.	1 h

Day 3 — Figures, prose, and the release gate

Roughly 6.5 hours, with the largest single risk sitting in 3.1.

#	Task	Done when	Est
3.1	The six unproduced figures. These are compiled into the paper and no script in the repository generates them: <code>gradient_flow.png</code> , <code>model_construction.png</code> , <code>rgrad_mumax_julia.png</code> , <code>true_hessian_julia.png</code> , <code>rgrad_walk.pdf</code> , <code>stoch_example.pdf</code> . Recommendation: do not write producers inside three days. Add one line to the reproducibility subsection stating that these six are drawn by hand and not generated, which is honest and costs minutes. Writing six producers is a day on its own and would put the deadline at risk.	Either producers exist, or the manuscript says plainly that they are hand-made.	2 h
3.2	Decide on the campaign figures. Not one <code>figs/exp*</code> appears in the manuscript — the entire benchmark figure output is currently outside the paper, which reads oddly for an experimental companion. Either wire the strongest few in (exp1 performance profiles, exp5 inactivity, exp8's countdown panel, which is the sharpest single picture the suite produces) or state that the tables carry the argument.	A deliberate decision, recorded.	1.5 h
3.3	Full read-through and provisional sweep. Nineteen markers at the start of Day 1; target zero. Any that survive should be converted from a marker into an explicit sentence saying what has not been measured — a reader can accept a stated limitation and cannot accept a red flag.	<code>grep -c provisional</code> returns only the definition.	2 h
3.4	Release gate. Clean build from an empty aux; confirm 0 undefined refs and 0 undefined citations; check the single overfull hbox; regenerate the archived PDF so it is no longer older than the source (it is currently 2 h behind).	PDF newer than <code>.tex</code> ; log clean; commit.	1 h

What could eat the budget

Risk	Cost if it lands	Mitigation
Writing six figure producers	+1 day	Declare them hand-made (3.1). This is the single biggest schedule risk.
exp11 CIFAR leg	+4.1 h	Excluded by default. Measured, not guessed: 99.3% of it is one model.
<code>true_stop = :both</code>	+1 h compute	<code>:test</code> costs nothing per iteration and refuses the bad stop; <code>:both</code> adds the true gradient.
Flat-well proposition needs new theory	+0.5 day	The numbers are stable and reproduced; if the statement resists, report the measurement as an observation rather than a proposition.
Numbers drift again mid-write	+2 h rework	Freeze the archives at the start of Day 2 and cite archive directory names in captions.

Verification commands

Each of these was run in producing this roadmap.

```
cd report && latexmk -pdf -interaction=nonstopmode Survey-Part-III-Numerical-v4.tex
grep -c "undefined" Survey-Part-III-Numerical-v4.log
julia --project=. -e 'using Pkg; Pkg.test()'
julia --project=docs docs/make.jl
julia --project=benchmark -e 'include("benchmark/initialisation.jl"); sampling_examples()'
julia --project=benchmark -e 'include("benchmark/initialisation.jl"); flat_well_experiment()'
```

Two notes on judgement. The day boundaries assume roughly six and a half working hours and no interruption; the ordering inside each day matters more than the boundaries, and task 1.1 in particular gates everything else. And the estimates for redaction tasks (1.5, 2.2, 3.3) are the least reliable numbers here — they are prose, not compute, and I have measured the compute far more directly than I can estimate the writing.